常見方法
Element Software

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常見方法

常用方法是用來擷取儲存叢集、API本身或持續API作業的相關資訊。

- GetAPI
- Get非 同步結果
- GetCompleteStats
- GetLimits
- GetOrigin
- GetRawStats
- 列表同步結果

如需詳細資訊、請參閱

- "NetApp SolidFire 資源頁面"
- "先前版本的NetApp SolidFire 產品及元素產品文件"

GetAPI

您可以使用「GetAPI」方法、取得系統中可使用的所有API方法和支援的API端點清單。

參數

此方法沒有輸入參數。

傳回值

此方法具有下列傳回值：

名稱	說明	類型
<版本>	此軟體版本所有支援的API方法清單、其中<版本>是此系統目前執行的軟體版本。	字串陣列
當前版本	儲存叢集軟體的目前版本。	字串
支援的Verions	系統支援的所有API端點清單。	字串陣列

申請範例

此方法的要求類似於下列範例：

```
{
  "method": "GetAPI",
  "params": {},
  "id" : 1
}
```

回應範例

此方法會傳回類似下列範例的回應：

```
{
"id": 1,
    "result": {
        "12.0": [
            "AbortSnapMirrorRelationship",
            "AddAccount",
            "AddClusterAdmin",
            "AddDrives",
            "AddIdpClusterAdmin",
            "AddInitiatorsToVolumeAccessGroup",
            "AddKeyServerToProviderKmip",
            "AddLdapClusterAdmin",
            "AddNodes",
            "AddVirtualNetwork",
            "AddVolumesToVolumeAccessGroup",
            "BreakSnapMirrorRelationship",
            "BreakSnapMirrorVolume",
            "CancelClone",
            "CancelGroupClone",
            "CheckPingOnVlan",
            "CheckProposedCluster",
            "CheckProposedNodeAdditions",
            "ClearClusterFaults",
            "CloneMultipleVolumes",
            "CloneVolume",
            "CompleteClusterPairing",
            "CompleteVolumePairing",
            "CopyVolume",
            "CreateBackupTarget",
            "CreateClusterInterfacePreference",
            "CreateClusterSupportBundle",
            "CreateGroupSnapshot",
            "CreateIdpConfiguration",
            "CreateInitiators",
```

```
            "CreateKeyProviderKmip",
            "CreateKeyServerKmip",
            "CreatePublicPrivateKeyPair",
            "CreateQoSPolicy",
            "CreateSchedule",
            "CreateSnapMirrorEndpoint",
            "CreateSnapMirrorEndpointUnmanaged",
            "CreateSnapMirrorRelationship",
            "CreateSnapMirrorVolume",
            "CreateSnapshot",
            "CreateStorageContainer",
            "CreateSupportBundle",
            "CreateVolume",
            "CreateVolumeAccessGroup",
            "DeleteAllSupportBundles",
            "DeleteAuthSession",
            "DeleteAuthSessionsByClusterAdmin",
            "DeleteAuthSessionsByUsername",
            "DeleteClusterInterfacePreference",
            "DeleteGroupSnapshot",
            "DeleteIdpConfiguration",
            "DeleteInitiators",
            "DeleteKeyProviderKmip",
            "DeleteKeyServerKmip",
            "DeleteQoSPolicy",
            "DeleteSnapMirrorEndpoints",
            "DeleteSnapMirrorRelationships",
            "DeleteSnapshot",
            "DeleteStorageContainers",
            "DeleteVolume",
            "DeleteVolumeAccessGroup",
            "DeleteVolumes",
            "DisableAutoip",
            "DisableBmcColdReset",
            "DisableClusterSsh",
            "DisableEncryptionAtRest",
            "DisableIdpAuthentication",
            "DisableLdapAuthentication",
            "DisableSnmp",
            "EnableAutoip",
            "EnableBmcColdReset",
            "EnableClusterSsh",
            "EnableEncryptionAtRest",
            "EnableFeature",
            "EnableIdpAuthentication",
            "EnableLdapAuthentication",
```

```
            "EnableSnmp",
            "GetAccountByID",
            "GetAccountByName",
            "GetAccountEfficiency",
            "GetActiveTlsCiphers",
            "GetAsyncResult",
            "GetBackupTarget",
            "GetBinAssignmentProperties",
            "GetClientCertificateSignRequest",
            "GetClusterCapacity",
            "GetClusterConfig",
            "GetClusterFullThreshold",
            "GetClusterHardwareInfo",
            "GetClusterInfo",
            "GetClusterInterfacePreference",
            "GetClusterMasterNodeID",
            "GetClusterSshInfo",
            "GetClusterState",
            "GetClusterStats",
            "GetClusterStructure",
            "GetClusterVersionInfo",
            "GetCompleteStats",
            "GetConfig",
            "GetCurrentClusterAdmin",
            "GetDefaultQoS",
            "GetDriveHardwareInfo",
            "GetDriveStats",
            "GetFeatureStatus",
            "GetFipsReport",
            "GetHardwareConfig",
            "GetHardwareInfo",
            "GetIdpAuthenticationState",
            "GetIpmiConfig",
            "GetIpmiInfo",
            "GetKeyProviderKmip",
            "GetKeyServerKmip",
            "GetLdapConfiguration",
            "GetLimits",
            "GetLldpInfo",
            "GetLoginBanner",
            "GetLoginSessionInfo",
            "GetNetworkConfig",
            "GetNetworkInterface",
            "GetNodeFipsDrivesReport",
            "GetNodeHardwareInfo",
            "GetNodeStats",
```

```
            "GetNtpInfo",
            "GetNvramInfo",
            "GetOntapVersionInfo",
            "GetOrigin",
            "GetPendingOperation",
            "GetProtectionDomainLayout",
            "GetQoSPolicy",
            "GetRawStats",
            "GetRemoteLoggingHosts",
            "GetSSLCertificate",
            "GetSchedule",
            "GetSnapMirrorClusterIdentity",
            "GetSnmpACL",
            "GetSnmpInfo",
            "GetSnmpState",
            "GetSnmpTrapInfo",
            "GetStorageContainerEfficiency",
            "GetSupportedTlsCiphers",
            "GetSystemStatus",
            "GetVirtualVolumeCount",
            "GetVolumeAccessGroupEfficiency",
            "GetVolumeAccessGroupLunAssignments",
            "GetVolumeCount",
            "GetVolumeEfficiency",
            "GetVolumeStats",
            "InitializeSnapMirrorRelationship",
            "ListAccounts",
            "ListActiveAuthSessions",
            "ListActiveNodes",
            "ListActivePairedVolumes",
            "ListActiveVolumes",
            "ListAllNodes",
            "ListAsyncResults",
            "ListAuthSessionsByClusterAdmin",
            "ListAuthSessionsByUsername",
            "ListBackupTargets",
            "ListBulkVolumeJobs",
            "ListClusterAdmins",
            "ListClusterFaults",
            "ListClusterInterfacePreferences",
            "ListClusterPairs",
            "ListDeletedVolumes",
            "ListDriveHardware",
            "ListDriveStats",
            "ListDrives",
            "ListEvents",
```

```
            "ListFibreChannelPortInfo",
            "ListFibreChannelSessions",
            "ListGroupSnapshots",
            "ListISCSISessions",
            "ListIdpConfigurations",
            "ListInitiators",
            "ListKeyProvidersKmip",
            "ListKeyServersKmip",
            "ListNetworkInterfaces",
            "ListNodeFibreChannelPortInfo",
            "ListNodeStats",
            "ListPendingActiveNodes",
            "ListPendingNodes",
            "ListProtectionDomainLevels",
            "ListProtocolEndpoints",
            "ListQoSPolicies",
            "ListSchedules",
            "ListServices",
            "ListSnapMirrorAggregates",
            "ListSnapMirrorEndpoints",
            "ListSnapMirrorLuns",
            "ListSnapMirrorNetworkInterfaces",
            "ListSnapMirrorNodes",
            "ListSnapMirrorPolicies",
            "ListSnapMirrorRelationships",
            "ListSnapMirrorSchedules",
            "ListSnapMirrorVolumes",
            "ListSnapMirrorVservers",
            "ListSnapshots",
            "ListStorageContainers",
            "ListSyncJobs",
            "ListTests",
            "ListUtilities",
            "ListVirtualNetworks",
            "ListVirtualVolumeBindings",
            "ListVirtualVolumeHosts",
            "ListVirtualVolumeTasks",
            "ListVirtualVolumes",
            "ListVolumeAccessGroups",
            "ListVolumeStats",
            "ListVolumeStatsByAccount",
            "ListVolumeStatsByVirtualVolume",
            "ListVolumeStatsByVolume",
            "ListVolumeStatsByVolumeAccessGroup",
            "ListVolumes",
            "ListVolumesForAccount",
```

```
            "ModifyAccount",
            "ModifyBackupTarget",
            "ModifyClusterAdmin",
            "ModifyClusterFullThreshold",
            "ModifyClusterInterfacePreference",
            "ModifyGroupSnapshot",
            "ModifyInitiators",
            "ModifyKeyServerKmip",
            "ModifyQoSPolicy",
            "ModifySchedule",
            "ModifySnapMirrorEndpoint",
            "ModifySnapMirrorEndpointUnmanaged",
            "ModifySnapMirrorRelationship",
            "ModifySnapshot",
            "ModifyStorageContainer",
            "ModifyVirtualNetwork",
            "ModifyVolume",
            "ModifyVolumeAccessGroup",
            "ModifyVolumeAccessGroupLunAssignments",
            "ModifyVolumePair",
            "ModifyVolumes",
            "PurgeDeletedVolume",
            "PurgeDeletedVolumes",
            "QuiesceSnapMirrorRelationship",
            "RemoveAccount",
            "RemoveBackupTarget",
            "RemoveClusterAdmin",
            "RemoveClusterPair",
            "RemoveDrives",
            "RemoveInitiatorsFromVolumeAccessGroup",
            "RemoveKeyServerFromProviderKmip",
            "RemoveNodes",
            "RemoveSSLCertificate",
            "RemoveVirtualNetwork",
            "RemoveVolumePair",
            "RemoveVolumesFromVolumeAccessGroup",
            "ResetDrives",
            "ResetNetworkConfig",
            "ResetNode",
            "ResetSupplementalTlsCiphers",
            "RestartNetworking",
            "RestartServices",
            "RestoreDeletedVolume",
            "ResumeSnapMirrorRelationship",
            "ResyncSnapMirrorRelationship",
            "RollbackToGroupSnapshot",
```

```
            "RollbackToSnapshot",
            "SecureEraseDrives",
            "SetClusterConfig",
            "SetClusterStructure",
            "SetConfig",
            "SetDefaultQoS",
            "SetLoginBanner",
            "SetLoginSessionInfo",
            "SetNetworkConfig",
            "SetNtpInfo",
            "SetProtectionDomainLayout",
            "SetRemoteLoggingHosts",
            "SetSSLCertificate",
            "SetSnmpACL",
            "SetSnmpInfo",
            "SetSnmpTrapInfo",
            "SetSupplementalTlsCiphers",
            "Shutdown",
            "SnmpSendTestTraps",
            "StartBulkVolumeRead",
            "StartBulkVolumeWrite",
            "StartClusterPairing",
            "StartVolumePairing",
            "TestAddressAvailability",
            "TestConnectEnsemble",
            "TestConnectMvip",
            "TestConnectSvip",
            "TestDrives",
            "TestHardwareConfig",
            "TestKeyProviderKmip",
            "TestKeyServerKmip",
            "TestLdapAuthentication",
            "TestLocalConnectivity",
            "TestLocateCluster",
            "TestNetworkConfig",
            "TestPing",
            "TestRemoteConnectivity",
            "UpdateBulkVolumeStatus",
            "UpdateIdpConfiguration",
            "UpdateSnapMirrorRelationship"
        ],
        "currentVersion": "12.0",
        "supportedVersions": [
            "1.0",
            "2.0",
            "3.0",
```

```
            "4.0",
            "5.0",
            "5.1",
            "6.0",
            "7.0",
            "7.1",
            "7.2",
            "7.3",
            "7.4",
            "8.0",
            "8.1",
            "8.2",
            "8.3",
            "8.4",
            "8.5",
            "8.6",
            "8.7",
            "9.0",
            "9.1",
            "9.2",
            "9.3",
            "9.4",
            "9.5",
            "9.6",
            "10.0",
            "10.1",
            "10.2",
            "10.3",
            "10.4",
            "10.5",
            "10.6",
            "10.7",
            "11.0",
            "11.1",
            "11.3",
            "11.5",
            "11.7",
            "11.8",
            "12.0"
        ]
    }
}
```

Get非 同步結果

您可以使用「Getynresult」擷取非同步方法呼叫的結果。有些方法呼叫需要一段時間才能執行、而且系統傳送初始回應時可能尚未完成。若要取得方法呼叫的狀態或結果、請使用「Getynresult」來輪詢方法傳回的asyncdyle值。

「GetSuccession Result」會以標準方式傳回作業的整體狀態（進行中、已完成或錯誤）、但針對作業傳回的實際資料則取決於原始方法呼叫、而且每種方法都會記錄傳回資料。

如果缺少keepResult參數或參數為假、則傳回結果時asyncdyle會變成非作用中狀態、之後會嘗試查詢該asyncdle傳回錯誤。您可以將keepResult參數設定為true、讓asyncdyle在未來的查詢中保持作用狀態。

參數

此方法具有下列輸入參數：

名稱	說明	類型	預設值	必要
asyncdle	從原始非同步方法呼叫傳回的值。	整數	無	是的
keepResult	如果為true、Getynpre Result在傳回時不會移除非同步結果、因此日後會對該asyncdle進行查詢。	布林值	錯	否

傳回值

此方法具有下列傳回值：

名稱	說明	類型
狀態	非同步方法呼叫的狀態。可能值： • 執行中：方法仍在執行中。 • 完成：方法已完成、且有結果或錯誤可供使用。	字串
結果	如果非同步方法成功完成、這是非同步作業的結果。如果非同步作業失敗、則此成員不存在。	字串
錯誤	如果狀態為「完整」、且非同步方法失敗、則此成員會包含錯誤詳細資料。如果非同步作業成功、則此成員不存在。	字串

名稱	說明	類型
結果類型	非同步方法呼叫正在執行或正在執行的操作類型。	字串
詳細資料	如果狀態為執行中、則此成員會包含與方法目前作業相關的資訊。如果非同步方法未執行、則此成員不存在。	Json物件
CREATETIM	以UTC +0格式呼叫非同步方法的時間。	ISO 8601日期字串
最後更新時間	非同步方法狀態上次更新的時間、格式為UTC+0。	ISO 8601日期字串

附註： GetSuccessynresult的傳回值基本上是標準Json回應的巢狀版本、並具有額外的狀態欄位。

申請範例

此方法的要求類似於下列範例：

```
{
  "method": "GetAsyncResult",
  "params": {
      "asyncHandle" : 389
},
"id" : 1
}
```

回應範例：方法錯誤

此方法會傳回類似下列範例的回應：

```
{
   "error": {
      "code": 500,
      "message": "DBClient operation requested on a non-existent path at
[/asyncresults/1]",
      "name": "xDBNoSuchPath"
   },
   "id": 1
}
```

如果「response」是來自Getynprynet呼叫的Json回應物件、則「response.error」會對應至Getynprynet方法本

身的錯誤（例如查詢不存在的asyncdyle）。

回應範例：非同步工作錯誤

此方法會傳回類似下列範例的回應：

```
{
   "id": 1,
   "result": {
     "createTime": "2016-01-01T02:05:53Z",
     "error": {
       "bvID": 1,
       "message": "Bulk volume job failed",
       "name": "xBulkVolumeScriptFailure",
       "volumeID": 34
     },
     "lastUpdateTime": "2016-01-21T02:06:56Z",
     "resultType": "BulkVolume",
     "status": "complete"
   }
}
```

「回應.result.error'」對應於原始方法呼叫所產生的錯誤。

回應範例：非同步工作成功

此方法會傳回類似下列範例的回應：

```
{
   "id": 1,
   "result": {
     "createTime": "2016-01-01T22:29:18Z",
     "lastUpdateTime": "2016-01-01T22:45:51Z",
     "result": {
       "cloneID": 25,
       "message": "Clone complete.",
       "volumeID": 47
     },
     "resultType": "Clone",
     "status": "complete"
   }
}
```

如果通話成功完成、「respose・resresult.result」是原始方法通話的傳回值。

新的自版本

9.6

GetCompleteStats

NetApp工程部門使用「GetCompleteStats」API方法來測試新功能。從「GetCompleteStats」傳回的資料不會記錄下來、經常變更、而且無法保證準確無誤。您不應使用「GetCompleteStats」來收集效能資料、或是與執行Element軟體的儲存叢集進行任何其他管理整合。

使用下列支援的API方法擷取統計資訊：

- GetVolume統計資料
- GetClusterStats
- GetNodeStats
- GetDriveStats

新的自版本

9.6

GetLimits

您可以使用「GetLimits」方法來取得API設定的限制值。這些值可能會在元素的版本之間變更、但若未更新系統、則不會變更。瞭解API設定的限制值、在撰寫面向使用者工具的API指令碼時非常實用。

> 無論用於傳遞方法的API端點版本為何、「GetLimits」方法都會傳回目前軟體版本的限制。

參數

此方法沒有輸入參數。

傳回值

此方法會傳回含有API限制之名稱值配對的Json物件。

申請範例

此方法的要求類似於下列範例：

```
{
   "method": "GetLimits",
   "id" : 1
}
```

回應範例

此方法會傳回類似下列範例的回應：

```
{
    "id": 1,
    "result": {
        "accountCountMax": 5000,
        "accountNameLengthMax": 64,
        "accountNameLengthMin": 1,
        "backupTargetNameLengthMax": 64,
        "backupTargetNameLengthMin": 1,
        "bulkVolumeJobsPerNodeMax": 8,
        "bulkVolumeJobsPerVolumeMax": 2,
        "chapCredentialsCountMax": 15000,
        "cloneJobsPerNodeMax": 8,
        "cloneJobsPerVirtualVolumeMax": 8,
        "cloneJobsPerVolumeMax": 2,
        "clusterAdminAccountMax": 5000,
        "clusterAdminInfoNameLengthMax": 1024,
        "clusterAdminInfoNameLengthMin": 1,
        "clusterPairsCountMax": 4,
        "fibreChannelVolumeAccessMax": 16384,
        "initiatorAliasLengthMax": 224,
        "initiatorCountMax": 10000,
        "initiatorNameLengthMax": 224,
        "initiatorsPerVolumeAccessGroupCountMax": 128,
        "iscsiSessionsFromFibreChannelNodesMax": 4096,
        "maxAuthSessionsForCluster": 1024,
        "maxAuthSessionsPerUser": 1024,
        "nodesPerClusterCountMax": 100,
        "nodesPerClusterCountMin": 3,
        "qosPolicyCountMax": 500,
        "qosPolicyNameLengthMax": 64,
        "qosPolicyNameLengthMin": 1,
        "scheduleNameLengthMax": 244,
        "secretLengthMax": 16,
        "secretLengthMin": 12,
        "snapMirrorEndpointIPAddressesCountMax": 64,
```

```
        "snapMirrorEndpointsCountMax": 4,
        "snapMirrorLabelLengthMax": 31,
        "snapMirrorObjectAttributeValueInfoCountMax": 9900000,
        "snapshotNameLengthMax": 255,
        "snapshotsPerVolumeMax": 32,
        "storageNodesPerClusterCountMin": 2,
        "virtualVolumeCountMax": 8000,
        "virtualVolumesPerAccountCountMax": 10000,
        "volumeAccessGroupCountMax": 1000,
        "volumeAccessGroupLunMax": 16383,
        "volumeAccessGroupNameLengthMax": 64,
        "volumeAccessGroupNameLengthMin": 1,
        "volumeAccessGroupsPerInitiatorCountMax": 1,
        "volumeAccessGroupsPerVolumeCountMax": 64,
        "volumeBurstIOPSMax": 200000,
        "volumeBurstIOPSMin": 100,
        "volumeCountMax": 4000,
        "volumeMaxIOPSMax": 200000,
        "volumeMaxIOPSMin": 100,
        "volumeMinIOPSMax": 15000,
        "volumeMinIOPSMin": 50,
        "volumeNameLengthMax": 64,
        "volumeNameLengthMin": 1,
        "volumeSizeMax": 17592186044416,
        "volumeSizeMin": 1000000000,
        "volumesPerAccountCountMax": 2000,
        "volumesPerGroupSnapshotMax": 32,
        "volumesPerVolumeAccessGroupCountMax": 2000,
        "witnessNodesPerClusterCountMax": 4
    }
}
```


新的自版本

9.6

GetOrigin

您可以使用「GetOrigin」方法取得節點建置位置的原始憑證。

參數

> 如果沒有來源認證、此方法會傳回「null」。

此方法沒有輸入參數。

傳回值

此方法會傳回廠商的認證資訊。

申請範例

此方法的要求類似於下列範例：

```
{
 "method": "GetOrigin",
 "id" : 1
}
```

回應範例

此方法會傳回類似下列範例的回應：

```
{
  "integrator": "SolidFire",
  "<signature>": {
    "pubkey": [public key info],
    "version": 1,
    "data": [signature info]
  },
   "contract-id": "none",
   "location": "Boulder, CO",
   "organization": "Engineering",
   "type": "element-x"
  }
 ]
}
```

新的自版本

9.6

GetRawStats

NetApp工程部門使用「GetRawStats」API方法來測試新功能。從「GetRawStats」傳回的資料不會記錄下來、經常變更、而且無法保證準確無誤。您不應使用「GetRawStats」來收集效能資料、或是與執行Element軟體的儲存叢集進行任何其他管理整合。

使用下列支援的API方法擷取統計資訊：

- GetVolume統計資料
- GetClusterStats
- GetNodeStats
- GetDriveStats

新的自版本

9.6

列表同步結果

您可以使用「list非 同步結果」來列出系統上所有目前執行和完成的非同步方法結果。使用「listynch Results」查詢非同步結果並不會導致完成同步處理的到期、您可以使用「Getynprynresult」查詢「listynch Results」傳回的任何同步處理。

參數

此方法具有下列輸入參數：

名稱	說明	類型	預設值	必要
asyncdResultTypes	結果類型的選用清單。您可以使用此清單、將結果限制為只有這些類型的作業。可能值： • DriveAdd：涉及系統的作業、將磁碟機新增至叢集。 • BulkVolume：在磁碟區之間進行複製作業、例如備份或還原。 • Clone：Volume Cloning作業。 • 磁碟機移除：系統從磁碟機複製資料以準備將資料從叢集移除的作業。 • RtfiPendingNode：系統在將相容軟體新增至叢集之前、先在節點上安裝相容軟體的相關作業。	字串陣列	無	否

傳回值

此方法具有下列傳回值：

名稱	說明	類型
asyncdynles	序列化非同步方法結果陣列。	Json物件陣列

申請範例

此方法的要求類似於下列範例：

```
{
   "method": "ListAsyncResults",
   "params": {
   },
   "id": 1
}
```

回應範例

此方法會傳回類似下列範例的回應：

```
{
   "id": 1,
   "result": {
     "asyncHandles": [
       {
         "asyncResultID": 47,
         "completed": true,
         "createTime": "2016-01-01T22:29:19Z",
         "data": {
           "cloneID": 26,
           "message": "Clone complete.",
           "volumeID": 48
         },
         "lastUpdateTime": "2016-01-01T22:45:43Z",
         "resultType": "Clone",
         "success": true
      },
      ...]
   }
}
```

新的自版本

9.6

如需詳細資訊、請參閱

Get非 同步結果

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